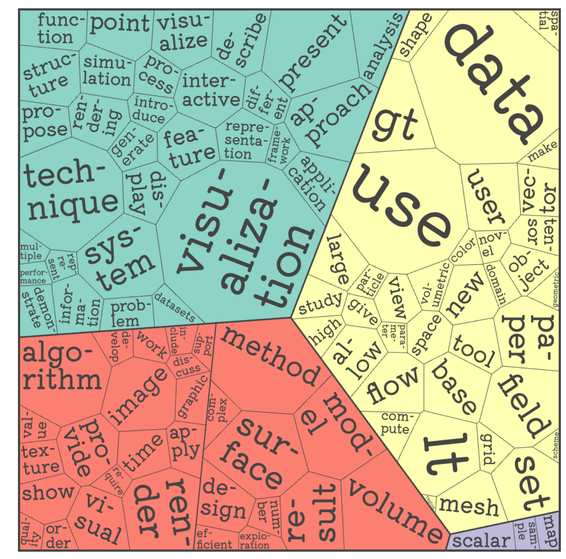
Story Gem| Extending Voronoi Treemap-Based Text Visualization to Non-Convex Regions

Naoya Oda, Yosuke Onoue
Nihon University

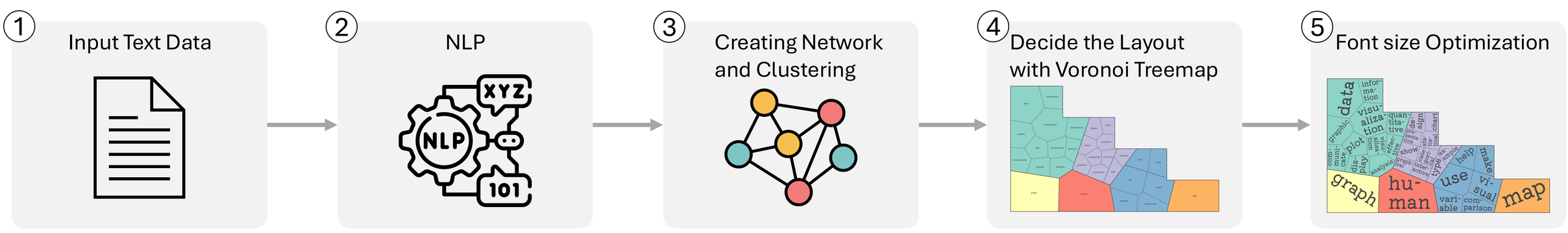
Introduction

WordCloud is a widely used text visualization technique that efficiently fills a region with words of varying sizes.
However, it places words randomly, without preserving any semantic relationships among them.
SemanticWordCloud addresses this limitation by arranging words according to their semantic proximity, yet it struggles to pack words efficiently within a given region.
StoryGem successfully overcomes these challenges by combining semantic preservation with space-efficient word placement, achieving meaningful layouts that fill the region without wasted space.
Nevertheless, StoryGem is constrained by its requirement that all display regions must be convex, which limits its applicability in many real-world scenarios.

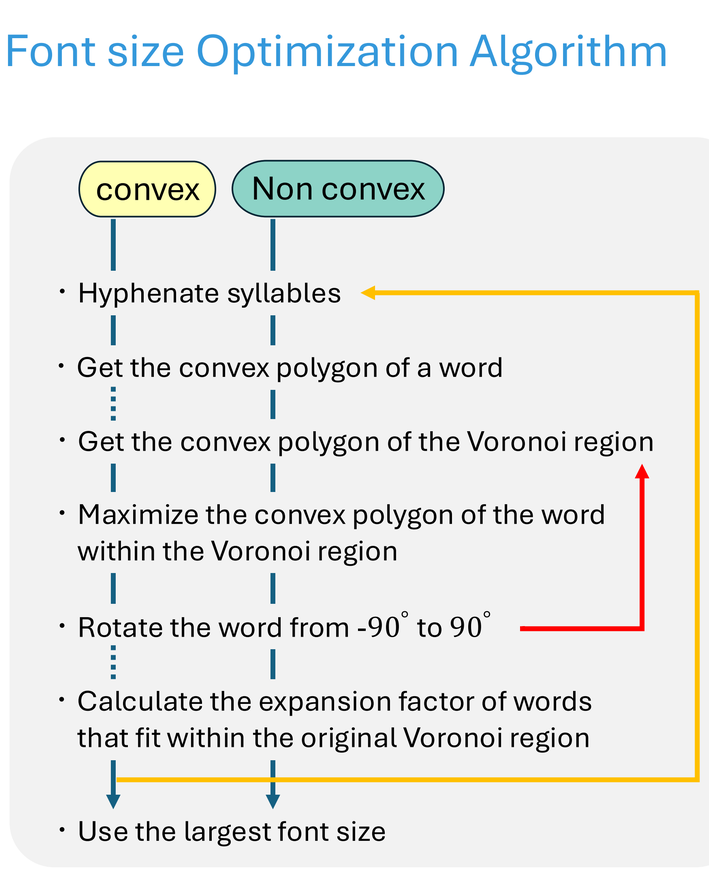


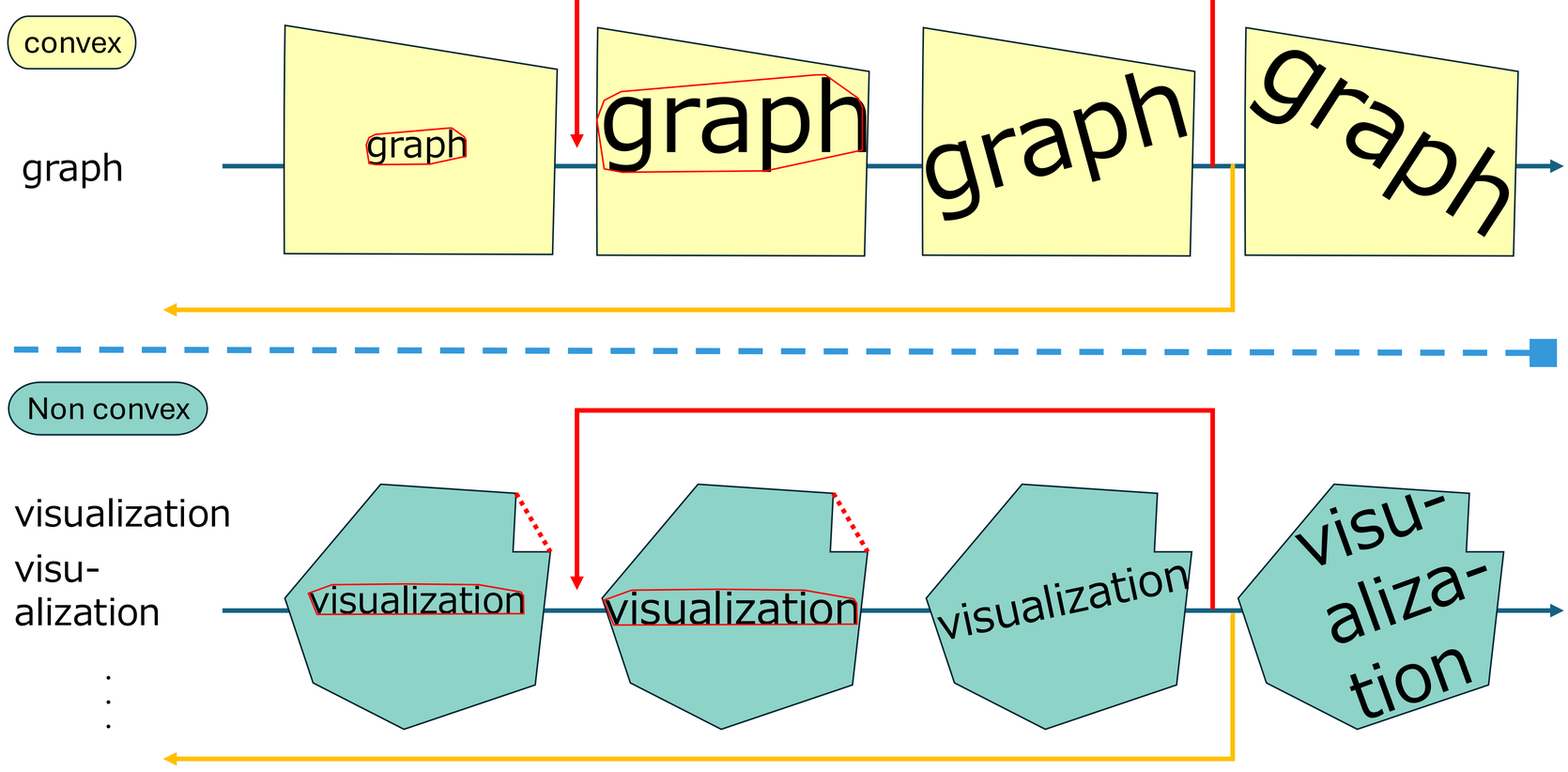
We propose an extension of StoryGem that enables its execution on non-convex regions, thereby broadening its utility for practical and contextually meaningful visualizations.

Method

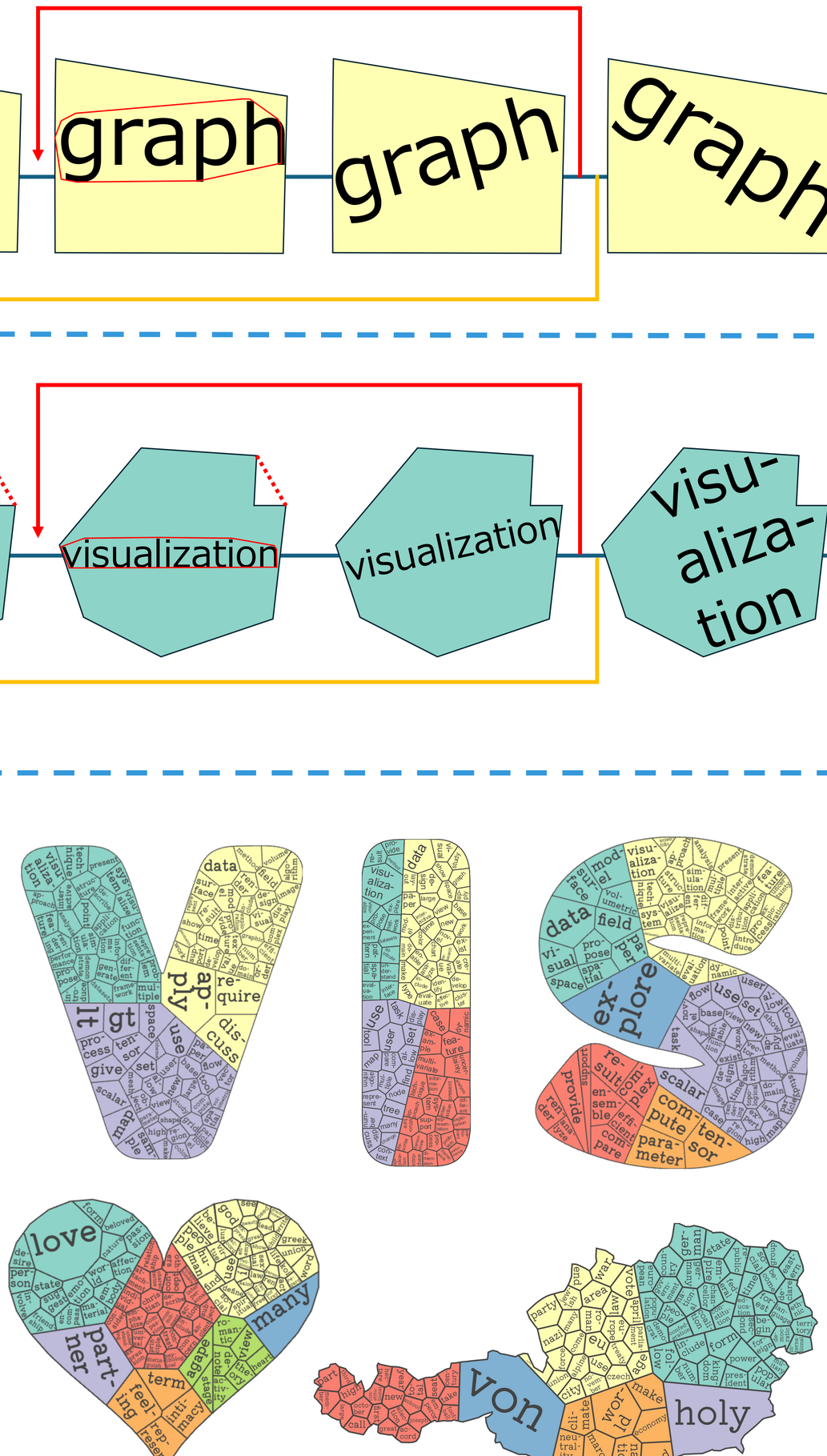


Font size Optimization Algorithm





Application Examples

To demonstrate the versatility of our extended StoryGem approach,
we applied it to a variety of semantically meaningful and non-convex shapes:

- **“V”, “I”, and “S” letter shapes**:
  Each letter was used as a region to visualize abstracts from the *VisPub Dataset* a collection of papers published in IEEE VIS.
  - “V” represents papers from the **InfoVis** track.
  - “I” represents papers from the **SciVis** track.
  - “S” represents papers from the **VAST** track.
- **Heart-shaped region**:
  The Wikipedia article on “love” was visualized within a heart-shaped region, showing how thematically relevant shapes can reinforce the meaning of the underlying text.
- **Austria map region**:
  Using the shape of Austria’s national boundary, we visualized the Wikipedia article on “Austria.” This example highlights the method’s applicability to geographic contexts, enabling regional text visualization aligned with actual map boundaries.